

#Atmospheric Dynamics



Chapter 3: The Equation Of Motion



Learning outcome

- *To Be acquainted with the Material derivative*
- *To Understand the Derivation of Equation of Motion*
- *To Perceive the Scale Analysis of the Equation Of Motion*
- *To Be familiar with Hydrostatic Equation*

Total Differentiation

Total Differentiation

- ❖ In order to apply conservation laws in the Eulerian frame, it is necessary to derive a relationship between the *rate of change of a field variable following the motion* and its *rate of change at a fixed point*.
- ❖ The rate of change of a field variable following the motion is called the *substantial*, *the total*, or *the material derivative* (it is denoted by D/Dt).
- ❖ *Material* derivative relates the Lagrangian and Eulerian description of flow.

Total Differentiation

- ❑ The rate of change of a field variable at a fixed point is called the *Eulerian* **local derivative** (*the partial derivative with respect to time*).
- ❑ It represents the rate of change of the material property introduced by temporal changes in the field variable at the position x where the material element is located.
- ❑ If the total derivative of a field variable is zero, then that variable is a conservative quantity following the motion

Total Differentiation

- ❑ To derive a relationship between the total derivative and the local derivative, let us refer to a particular field variable (e.g. temperature).
- ❑ The local change is then entirely due to advection
- ❑ Advection is the process of transport of a substance or property (such as heat, moisture, or pollutants) by the bulk motion (the overall, coherent movement of a large volume of fluid as a whole, rather than the random, individual movement of particles within the fluid)

Total Differentiation

- In order to relate the total derivative to the local rate of change at a fixed point, we consider the temperature measured on a balloon that moves with the wind.
- Suppose that this temperature is T_0 at the point x_0, y_0, z_0 and time t_0 . If the balloon moves to the point $x_0 + \delta x, y_0 + \delta y, z_0 + \delta z$ in a time increment δt , then the temperature change recorded on the balloon, δT , can be expressed in a Taylor series expansion as

$$\delta T = \left(\frac{\partial T}{\partial t} \right) \delta t + \left(\frac{\partial T}{\partial x} \right) \delta x + \left(\frac{\partial T}{\partial y} \right) \delta y + \left(\frac{\partial T}{\partial z} \right) \delta z + (\text{higher order terms})$$

Total Differentiation

□ Dividing through by δt and noting that δT is the change in temperature following the motion so that

$$\frac{DT}{Dt} \equiv \lim_{\delta t \rightarrow 0} \frac{\delta T}{\delta t}$$

in the limit $\delta t \rightarrow 0$

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + \left(\frac{\partial T}{\partial x} \right) \frac{Dx}{Dt} + \left(\frac{\partial T}{\partial y} \right) \frac{Dy}{Dt} + \left(\frac{\partial T}{\partial z} \right) \frac{Dz}{Dt}$$

$$\frac{Dx}{Dt} \equiv u, \frac{Dy}{Dt} \equiv v, \frac{Dz}{Dt} \equiv w$$

Total Differentiation

- Then u , v , w are the velocity components in the x , y , z directions, respectively, and

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right)$$

- Using vector notation this expression may be rewritten as

$$\mathbf{U} = \mathbf{i}u + \mathbf{j}v + \mathbf{k}w$$

$$\frac{\partial T}{\partial t} = \frac{DT}{Dt} - \mathbf{U} \cdot \nabla T$$

Advection term

- The local rate of change of temperature equals the rate of change of temperature following the motion (i.e., the heating or cooling of individual air parcels) plus the advective rate of change of temperature.
- This relationship holds for any other fields

□ For example, we may wish to relate the pressure change measured by a barometer on a moving ship to the local pressure change.

Exercise

□ The surface pressure decreases by 3 hPa per 180 km in the eastward direction. A ship steaming eastward at 10 km/h measures a pressure fall of 1 hPa per 3 h. What is the pressure change on an island that the ship is passing?

Equation of Motion

Equation of Motion

- ❑ Forces cause acceleration. When there are no forces, or when all the forces cancel each other, there is no acceleration.
- ❑ We have seen that the primary forces in the *vertical direction* are *gravity and the centrifugal force* acting downward and the pressure gradient force acting upward.
- ❑ There is also a small vertical component in the Coriolis force.
- ❑ The forces acting in the *horizontal direction* are the *pressure gradient force, the viscous force, and the Coriolis force*

Equation of Motion

□ We can use this information to construct an equation that calculates the net force acting on a parcel of air, and hence its acceleration.

□ First step in deriving the equation of motion for a fluid element in the atmosphere is Apply Newton's second law to the fluid element.

□ If we can perform this calculation at every point in the atmosphere, then, in theory, we can determine the winds anywhere on the Earth.

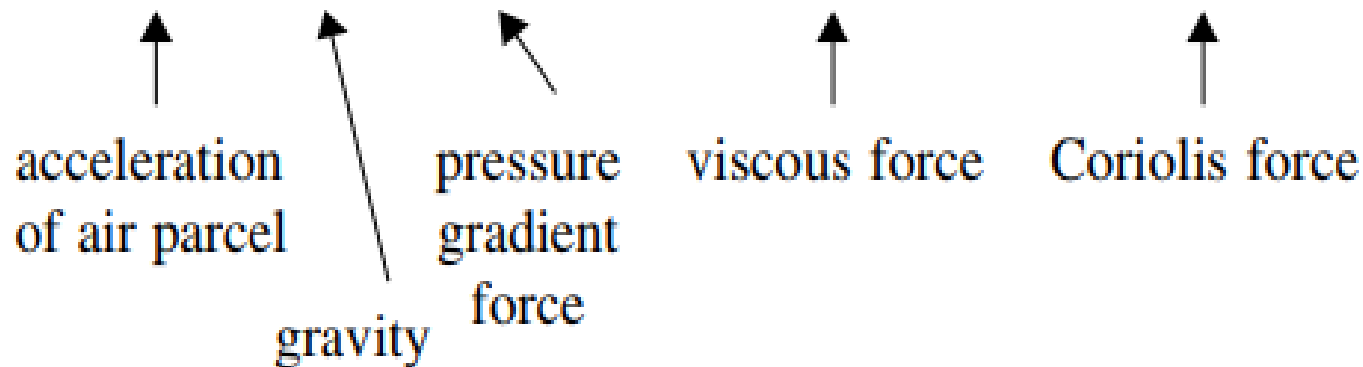
□ To construct the equation, we simply sum all of the forces acting in each vector direction, and equate this sum to the acceleration of the parcel.

Equation of Motion

$$\frac{Du}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + (2\Omega v \sin \phi - 2\Omega w \cos \phi)$$

$$\frac{Dv}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) - (2\Omega u \sin \phi)$$

$$\frac{Dw}{Dt} = -g - \frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + (2\Omega u \cos \phi)$$



- ✓ This set of equations is usually called the **Navier–Stokes equations** for the conservation of momentum.

Equation of Motion

- ❖ We can write the set of equations in some alternative forms. First, since we expect to be solving these equations in an Eulerian framework, it is often appropriate to replace the material derivative with the local derivative.

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + (2\Omega v \sin \phi - 2\Omega w \cos \phi)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) - (2\Omega u \sin \phi)$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -g - \frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + (2\Omega u \cos \phi)$$

SCALE ANALYSIS OF THE EQUATIONS OF MOTION

Scale Analysis:

- ❑ It is process of simplifying the equations of motion using typical scales for the phenomenon of interest
- ❑ It Simplifies equations by comparing the relative magnitudes of terms for specific atmospheric phenomena.
- ❑ It reduce the complexity of equations by identifying dominant terms
- ❑ Key scales: length, time, velocity, and pressure.
- ❑ It Helps identify dominant forces in different motions (e.g., geostrophic flow or hydrostatic balance).

SCALE ANALYSIS OF THE EQUATIONS OF MOTION

- ❖ The process of simplifying the equations of motion using typical scales for the phenomenon of interest is called *scale analysis*.
- ❖ As an illustration of the technique, we will perform a scale analysis of the Navier–Stokes equations for a typical midlatitude system.
- ❖ However, first we will make one prior simplification of the equation by examining the properties of the Coriolis force term

$$f = 2\Omega \sin \phi$$

For scaling purposes, we use the value of f at 45° latitude:

$$f_0 = 2\Omega \sin 45^\circ = 1.03 \times 10^{-4}$$

SCALE ANALYSIS OF THE EQUATIONS OF MOTION

- ❑ Chapter one discussed the basic notion of scaling the equations of motion in order to determine whether some terms in the equations are negligible for motions of meteorological concern.
- ❑ Elimination of terms on scaling considerations not only has the advantage of simplifying the mathematics, but as shown in later chapters, the elimination of small terms in some cases has the very important property of completely eliminating or *filtering* an *unwanted type of motion*.
- ❑ The complete equations of motion describe all types and scales of atmospheric motions.

- ❑ The Navier–Stokes equations are highly *nonlinear*, and cannot be solved analytically without considerable simplification.
- ❑ A nonlinear differential equation is one that includes products of terms involving the dependent variable – in this case, the nonlinear terms are the advection terms,

$$u\partial u/\partial x \text{ and } v\partial u/\partial y$$

for example. In nonlinear systems, different solutions cannot be superposed (added together) to form new solutions.

- ❑ This makes solving the equations much more difficult than linear systems. Hence, our approach in this class will be to simplify the equations in order to solve them.
- ❑ Since much of the significant weather in middle latitudes is associated with cyclones and, to a lesser extent, anticyclones, we will derive a simpler form of the equation specifically for these systems.
- ❑ Typical values for the appropriate scales are shown

Table 5.2 Typical scales for a mid-latitude weather system

Scale	Symbol	Magnitude
Horizontal wind scale	U	10 m s^{-1}
Vertical wind scale	W	10^{-2} m s^{-1}
Horizontal length scale	L	10^6 m
Vertical length scale (depth of troposphere)	H	10^4 m
Time scale (L/U)	T	10^5 s
Kinematic viscosity	ν	$10^{-5} \text{ m}^2 \text{ s}^{-1}$
Dynamic pressure scale	$\delta p / \rho$	$10^3 \text{ m}^2 \text{ s}^{-2}$
Total pressure scale	P / ρ	$10^5 \text{ m}^2 \text{ s}^{-2}$
Gravity	g	10 m s^{-2}
Density variation scale	$\delta \rho / \rho$	10^{-2}

Table 5.2.

Then, the process of scale analysis is conducted as follows:

x-eqn	$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$	$= -\frac{1}{\rho} \frac{\partial p_d}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \nu \frac{\partial^2 u}{\partial z^2} + 2\Omega v \sin \phi - 2\Omega w \cos \phi$						
y-eqn	$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$	$= -\frac{1}{\rho} \frac{\partial p_d}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \nu \frac{\partial^2 v}{\partial z^2} - 2\Omega u \sin \phi$						
scale	$\frac{U^2}{L}$	$\frac{U^2}{L}$	$\frac{UW}{H}$	$\frac{\delta p}{\rho L}$	$\frac{\nu U}{L^2}$	$\frac{\nu U}{H^2}$	$f_0 U$	$\approx f_0 W$
magnitude	10^{-4}	10^{-4}	10^{-5}	10^{-3}	10^{-16}	10^{-12}	10^{-3}	10^{-6}

$$-fv \approx -\frac{1}{\rho} \frac{\partial p}{\partial x}; \quad fu \approx -\frac{1}{\rho} \frac{\partial p}{\partial y}$$

Geostrophic Approximation

It is apparent from Table 5.2 that for midlatitude synoptic scale disturbances the Coriolis force (term B) and the pressure gradient force (term F) are in approximate balance. Retaining only these two terms in (2.19) and (2.20) gives as a first approximation the *geostrophic* relationship

$$-fv \approx -\frac{1}{\rho} \frac{\partial p}{\partial x}; \quad fu \approx -\frac{1}{\rho} \frac{\partial p}{\partial y}$$

- ❑ Based on these magnitudes, we can see that the viscous term makes a very small contribution to the force balance experienced in a typical mid-latitude cyclone.
- ❑ So we can ignore that term with very little impact on the precision of our results.
- ❑ In fact, to a good degree of accuracy, we can also discard the Coriolis deflection caused by the vertical motion in the system , and the vertical advection term .
- ❑ Hence, we can rewrite the horizontal component of the Navier–Stokes

equations in

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p_d}{\partial x} + 2\Omega v \sin \phi$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p_d}{\partial y} - 2\Omega u \sin \phi$$

Table 2.2 *Scale Analysis of the Vertical Momentum Equation*

z - Eq.	Dw/Dt	$-2\Omega u \cos \phi$	$-(u^2 + v^2)/a$	$= -\rho^{-1} \partial p / \partial z$	$-g$	$+F_{rz}$
Scales	UW/L	$f_0 U$	U^2/a	$P_0/(\rho H)$	g	νWH^{-2}
m s^{-2}	10^{-7}	10^{-3}	10^{-5}	10	10	10^{-15}

$$\frac{1}{\rho_0} \frac{dp_0}{dz} \equiv -g$$

Hydrostatic Equation

- ❑ Hydrostatic equilibrium occurs when the force of gravity and the vertical pressure gradient have equal values and oppose each other

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Thanks !